0.0.1 Field Theory

Lemma 0.0.1. Suppose that $\varphi : F \to F'$ is a Field-Homomorphism. If φ is not zero map, then φ is injective.

Proof. Since φ is not zero map, $\varphi(a) \neq 0_{F'}$ for some $a \in F$. Thus, $\ker\varphi \subsetneq F$.
But, the field F has only ideal for $\{0\}$ or F, hence the ideal $\ker\varphi$ must be $\{0\}$. Now,

$$\varphi(a) = \varphi(b) \implies \varphi(a-b) = 0 \implies a - b = 0$$

□

Lemma 0.0.2. Suppose that F is a field.
If $p(x) \in F[x]$ is irreducible in $F[x]$, then $F[x]/(p(x))$ is a field.

Proof. Since F is a field, $F[x]$ is Euclidean domain, P.I.D., and U.F.D. Thus,

$$p(x) \text{ irreducible } \underset{\text{U.F.D.}}{\implies} (p(x)) \text{ prime } \underset{\text{P.I.D.}}{\implies} (p(x)) \text{ maximal in } F[x]$$

Hence, $K = F[x]/(p(x))$ becomes a field. □

Theorem 0.0.3. Suppose that F is a field, and $p(x) \in F[x]$ is irreducible.
Then, $K \overset{\text{def}}{=} F[x]/(p(x))$ is a field containing an isomorphic copy of F, and $p(x)$ has a root α in K.

Proof. Since F is a field, $F[x]$ is Euclidean domain, P.I.D., and U.F.D. Thus,

$$p(x) \text{ irreducible } \underset{\text{U.F.D.}}{\implies} (p(x)) \text{ prime } \underset{\text{P.I.D.}}{\implies} (p(x)) \text{ maximal in } F[x]$$

Hence, $K = F[x]/(p(x))$ becomes a field. Consider the Canonical projection

$$\pi : F[x] \to F[x]/(p(x)) : f(x) \mapsto f(x) + (p(x))$$

and restriction over F, $\varphi = \pi_F : F \to F[x]/(p(x))$ is Homomorphism with $\ker\varphi = \{0\}$, because $\varphi(1) + (p(x)) \neq (p(x))$. Now, $F \cong F/\ker\varphi \cong \varphi[F] \subseteq K = F[x]/(p(x))$. Further, denote $\overline{x} \overset{\text{def}}{=} \pi(x) = x + (p(x))$, and $p(x) = \sum_{i=0}^{n} a_i x^i$. Then,

$$p(\overline{x}) = \sum_{i=0}^{n} \overline{a_i}\overline{x}^i = \sum_{i=0}^{n} \pi(a_i)\pi(x)^i = \pi\left(\sum_{i=0}^{n} a_i x_i\right) = \pi(p(x)) = \overline{p(x)} = \overline{0} \in K$$

□

Theorem 0.0.4. Suppose that F is a field, and $p(x) \in F[x]$ is irreducible.
If K is an Extension field of F containing a root $\alpha \in K$ of $p(x)$, then $F(\alpha) \cong F[x]/(p(x))$.

Proof. Define a map $\varphi_\alpha : F[x]/(p(x)) \to F(\alpha) : f(x) + (p(x)) \mapsto f(\alpha)$.
Well-defined: Suppose that $f(x) + (p(x)) = g(x) + (p(x))$. Then, $\varphi_\alpha(f(x) + (p(x))) = f(\alpha)$ and $\varphi_\alpha(g(x) + (p(x))) = g(\alpha)$.

$$f(x) + (p(x)) = g(x) + (p(x)) \iff f(x) - g(x) \in (p(x))$$

$$f(x) - g(x) \in (p(x)) \implies f(x) - g(x) = p(x)r(x) \text{ for some } r(x) \in F[x] \implies f(\alpha) - g(\alpha) = p(\alpha)r(\alpha) = 0 \cdot r(\alpha) = 0.$$

Ring-Homomorphism:

1. $\varphi([f(x) + (p(x))] + [g(x) + (p(x))]) = \varphi(f(x) + g(x) + (p(x))) = f(\alpha) + g(\alpha) = \varphi(f(x) + (p(x))) + \varphi(g(x) + (p(x)))$.
2. $\varphi([f(x) + (p(x))] \cdot [g(x) + (p(x))]) = \varphi(f(x)g(x) + (p(x))) = f(\alpha)g(\alpha) = \varphi(f(x) + (p(x))) \cdot \varphi(g(x) + (p(x)))$.

Injectivity: Since φ is a non-trivial homomorphism from field into field, thus $\ker\varphi = \{0\}$.
(More precisely, $\varphi(1) = 1 + (p(x)) \neq (p(x))$ because $1 \notin (p(x))$. Thus, $\ker\varphi \subsetneq F[x]/(p(x)) \implies \ker\varphi = \{0\}$.)
Now, the φ becomes an Embedding:

$$\varphi[F[x]/(p(x))] \cong F[x]/(p(x))$$

Since $F[x]/(p(x))$ is a field, the image $\varphi[F[x]/(p(x))]$ is a field, being it is isomorphic copy.
The definition of $F(\alpha)$ gives $\varphi[F[x]/(p(x))] \supset F(\alpha)$, and $\varphi[F[x]/(p(x))] \subset F(\alpha)$ is trivial. Consequently,

$$F(\alpha) = \varphi[F[x]/(p(x))] \cong F[x]/(p(x))$$

□

Theorem 0.0.5. Suppose that $p(x) \in F[x]$ is an irreducible with $\deg p(x) = n$, and $K = F[x]/(p(x))$.
Put $\overline{x} = x + (p(x)) \in K$. Then, $\{1, \overline{x}, \overline{x}^2, \ldots, \overline{x}^{n-1}\}$ is basis for K as Vector space over F.

Proof. Above lemma guarantees $K = F[x]/(p(x))$ is a field.
1) *Linealy independent.*
Suppose that $a_0 + a_1\overline{x} + \cdots + a_{n-1}\overline{x}^{n-1} = 0$ in K where a_i are not all zero. Then,

$$\begin{aligned} 0 =& a_0 + a_1\overline{x} + \cdots + a_{n-1}\overline{x}^{n-1} \\ =& a_0 + a_1(x + (p(x))) + \cdots + a_{n-1}(x^{n-1} + (p(x))) \\ =& a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} + (p(x)) \\ \iff& a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} = p(x)b(x) \text{ for some } b(x) \in F[x] \end{aligned}$$

But, the left term has a degree at most $n-1$, and the right term has a degree at least n. Contradiction.
2) $F[x]$ *is generated by* $\{1, \overline{x}, \ldots, \overline{x}^{n-1}\}$.
Let $a(x) \in F[x]$. Then, the Euclidean algorithm for $F[x]$ gives that: there exist uniquely $q(x), r(x) \in F[x]$ s.t

$$a(x) = q(x)p(x) + r(x) \text{ with } \deg r(x) < \deg p(x)$$

Since $q(x)p(x) \in (p(x))$, $a(x) + (p(x)) = r(x) + (p(x))$, the degree of $a(x) + (p(x))$ is smaller than n.
Now, for any $a(x) + (p(x)) \in K = F[x]/(p(x))$, $a(x) + (p(x)) \in \langle\{1, \overline{x}, \cdots, \overline{x}^{n-1}\}\rangle$. □

Corollary 0.0.6. Suppose that F is a field, and $p(x) \in F[x]$ is irreducible polynomial with $\deg p(x) = n$.
If E is an Extension field of F containing α as a root of $p(x)$. Then,

$$F(\alpha) = \{a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{n-1}\alpha^{n-1} \mid a_i \in F\} \subseteq E$$

Proof. Since $F[x]/(p(x)) = \{a_0 + a_1\overline{x} + \cdots + a_{n-1}\overline{x}^{n-1} \mid a_i \in F\} \cong F(\alpha)$, being the isomorphsim

$$\varphi_\alpha : F[x]/(p(x)) \to F(\alpha) : f(x) + (p(x)) = \overline{f(x)} \mapsto f(\alpha)$$

Thus, combining with above fact,

$$\begin{aligned} F(\alpha) = \varphi_\alpha[F[x]/(p(x))] &= \{\varphi_\alpha(a_0 + a_1\overline{x} + \cdots + a_{n-1}\overline{x}^{n-1}) \mid a_i \in F\} \\ &= \{\varphi_\alpha(a_0) + \varphi_\alpha(a_1\overline{x}) + \cdots + \varphi_\alpha(a_{n-1}\overline{x}^{n-1}) \mid a_i \in F\} \\ &= \{a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{n-1}\alpha^{n-1} \mid a_i \in F\} \end{aligned}$$

□

0.0.2 Extension Field

Definition 0.0.7. Suppose that F is a field, and K be a field containing F as a subfield.
The element $\alpha \in K$ is called *algebraic* over F if:

$$\text{There exists a non-zero polynomial } f(x) \in F[x] \text{ such that } f(\alpha) = 0 \text{ in } K.$$

If not algebraic, it is called *transcendental* over F.

Theorem 0.0.8. Suppose that $\alpha \in K$ is algebraic over F.
Then, there exists a unique monic irreducible polynomial $m_{\alpha,F}(x) \in F[x]$ such that $m_{\alpha,F}(\alpha) = 0$. Moreover,

$$f(x) \in F[x] \text{ has } \alpha \text{ as a root } \textit{if and only if } m_{\alpha,F}(x) \in F[x] \text{ divides } f(x) \text{ in } F[x].$$

Proof. Since $\alpha \in K$ is algebraic over F, the Well-Ordering principle gives the existence of minimal degree polynomial $g(x) \in F[x]$ having α as a root. More precisely,

$$S = \{\deg f(x) \in \mathbb{N} \mid f(x) \in F[x] \text{ s.t } f(\alpha) = 0\}$$

is non-emtpy since α is algebraic. Therefore, the Well-Ordering Principle gives that S has a minimum $n \in \mathbb{N}$.
Now, we can choose $g(x) \in F[x]$ with $\deg g(x) = n$ and $g(\alpha) = 0$.
Using Contradiction: Suppose that $g(x)$ is reducible. That is, $g(x) = a(x)b(x)$ with $\deg a, \deg b < \deg g$.
Since $g(\alpha) = a(\alpha)b(\alpha) = 0$ and K is a field, either $a(\alpha)$ or $b(\alpha)$ must be zero.
This contradicts with $n \in \mathbb{N}$ is the minimum degree of polynomial which has α as a root.
In summary:

If $g(x) \in F[x]$ is a polynomial of minimal degree which has α as a root, then $g(x)$ is irreducible over F.

Now, Suppose that $f(x) \in F[x]$ is any polynomial having α as a root. Then, $\deg f(x) \geq \deg g(x)$ by setting.
By the Euclidean Algorithm in $F[x]$, there exist unique polynoimals $q(x), r(x) \in F[x]$ such that

$$f(x) = q(x)g(x) + r(x) \text{ with } \deg r(x) < \deg g(x)$$

Meanwhile,

$$0 = f(\alpha) = q(\alpha)g(\alpha) + r(\alpha) = q(\alpha) \cdot 0 + r(\alpha) = r(\alpha)$$

$r(\alpha) = 0$ implies $r(x)$ must be zero, by minimality. Hence, $g(x)$ divides $f(x)$ in $F[x]$.
Finally, put $a \in F$ be a leading coefficient of $g(x)$. Then, $m_{\alpha,F}(x) = a^{-1}g(x)$ is monic polynomial.
And, Uniqueness given by: for any $f(x) \in F[x], \ f(\alpha) = 0 \implies m_{\alpha,F}(x) \mid f(x)$. □

Corollary 0.0.9. If L is an Extension of field F and $\alpha \in L$ is algebraic over both F and L,
then $m_{\alpha,L}(x)$ divides $m_{\alpha,F}(x)$ in $L[x]$.

Above theorem allows defining:

Definition 0.0.10. Suppose that K is an Extension field of a field F, and $\alpha \in K$ is algebric over F.
The monic irreducible polynomial which has α as root is called *the minial polynomial* for α over F.
Denote this $m_{\alpha,F}(x) \in F[x]$ or $\mathtt{irr}(\alpha, F)$.

Proposition 0.0.11. Suppose that F is a field and E is an Extension of F.
If $\alpha \in E$ is algebraic over F, then $F(\alpha) \cong F[x]/(m_{\alpha,F}(x))$ and

$$[F(\alpha) : F] \stackrel{\text{def}}{=} \dim_F F(\alpha) = \deg m_{\alpha,F}(x) \stackrel{\text{def}}{=} \deg \alpha$$

Theorem 0.0.12. Suppose that E is an Extension field of a field F, and $\alpha \in E$.

$\alpha \in E$ is algebraic over F *if and only if* $[F(\alpha) : F]$ has finite dimension

Proof. If $\alpha \in E$ is algebraic over F, then $[F(\alpha) : F] = \deg m_{\alpha,F}(x)$, thus finite.
Precisely, if $f(x) \in F[x]$ satisfies $\deg f(x) = n$ and $f(\alpha) = 0$, then $[F(\alpha) : F] = \deg_F \alpha \leq n$.
Conversely, suppose $\alpha \in E$ where $[F(\alpha) : F] = n$. Then, the $n+1$ elements

$$1, \alpha, \alpha^2, \ldots, \alpha^n$$

must be Linearly dependent, thus for some not all zero elements $b_0, \ldots, b_n \in F$,

$$b_0 + b_1\alpha + b_2\alpha^2 + \cdots + b_n\alpha^n = 0$$

Now, the polynomial in $F[x]$

$$f(x) = b_0 + b_1x + b_2x^2 + \cdots + b_nx^n$$

satisfies $f(\alpha) = 0$ in $F(\alpha)$. In summary, if $[F(\alpha) : F] = n$, then $\alpha \in E$ is algebraic over F with $\deg_F \alpha \leq n$. □

Corollary 0.0.13. If E is an Extension field of F with $[E : F]$ has finite, then E is algebraic over F.

Theorem 0.0.14. Suppose that $F \subseteq L \subseteq E$ are fields. Then,

$$[E : F] = [E : L][L : F]$$

Proof. Suppose that $[E : L]$ and $[L : F]$ are finite. Put $[E : L] = m$ and $[L : F] = n$.
Set $A = \{\alpha_1, \alpha_2, \ldots, \alpha_m\} \subset E$ and $B = \{\beta_1, \beta_2, \ldots, \beta_n\} \subset L$ are basis of E over L and L over F, respectively.
Let $x \in E$ be given. Since A is basis for E over L, there exists a unique linear combination: for $a_i \in L$,

$$x = a_1\alpha_1 + a_2\alpha_2 + \cdots + a_m\alpha_m = \sum_{i=1}^{m} a_i\alpha_i$$

Meanwhile, for each $a_i \in L,\ 1 \leq i \leq m$, there exist unique linear combinations: for $b_{i,j} \in F$,

$$a_i = b_{i,1}\beta_1 + b_{i,2}\beta_2 + \cdots + b_{i,n}\beta_n = \sum_{j=1}^{n} b_{i,j}\beta_j$$

Combining above, we obtain

$$x = \sum_{i=1}^{m}\left(\sum_{j=1}^{n} b_{i,j}\beta_j\right)\alpha_i = \sum_{\substack{1\leq i\leq m\\1\leq j\leq n}} b_{i,j}\beta_j\alpha_i$$

Since each $b_{i,j} \in F$ and $\beta_j\alpha_i \in E$, the subset $C = \{\beta_j\alpha_i \mid \alpha_i \in A,\ \ \beta_j \in B\} \subset E$ spans E over F.
Meanwhile, this C is linearly independent over F, because: Suppose that

$$\sum_{i=1}^{m}\left(\sum_{j=1}^{n} c_{i,j}\beta_j\right)\alpha_i = 0$$

Since each $c_{i,j}\beta_j \in L$ and $A = \{\alpha_i\}$ is linearly independent over L, for each $1 \leq i \leq m$,

$$\sum_{j=1}^{n} c_{i,j}\beta_j = 0$$

$c_{i,j} \in F$ and $B = \{\beta_j\}$ is linearly independent over F, all $c_{i,j} = 0$. Thus, C is basis which has mn elements.
Meanwhile, if $[L : F]$ is infinite, then $[E : F]$ is clearly infinite; being $E \subset L$. Similarly, $[E : L]$. □

Theorem 0.0.15. Suppose that E is an algebraic Extension field of F. Then,

$$E \text{ is finite Extension} \iff \text{There exist finite elements } \alpha_1, \ldots, \alpha_n \in E \text{ such that } E = F(\alpha_1, \ldots, \alpha_n).$$

Proof. Suppose that E is finite Extension.
If $[E : F] = 1$, then $F(1) = F = E$, claim proved.
If $[E : F] > 1$, then $E \neq F$. Thus, put $\alpha_1 \in E \setminus F$. If $F(\alpha_1) = E$, then there is nothing to prove. If $F(\alpha_1) \neq E$, then put again $\alpha_2 \in E \setminus F(\alpha_1)$. Since $[E : F]$ is finite integer, this process will terminate in finite.
Thus for some $n \in \mathbb{N}$, $E = F(\alpha_1, \ldots, \alpha_n)$.
Conversely, suppose that $E = F(\alpha_1, \ldots, \alpha_n)$ for some $\alpha_i \in E$. Then,

$$E = [E : F(\alpha_n)][F(\alpha_n) : F(\alpha_{n-1})] \cdots [F(\alpha_1) : F]$$

Since E is algebraic, $[E : F(\alpha_n)], [F(\alpha_n) : F(\alpha_{n-1})], \ldots [F(\alpha_1) : F]$ are all finite. Thus $[E : F]$ is finite. □

Theorem 0.0.16. If E is Algebraic over K and K is Algebraic over F. Then, E is Algebraic over F.

Proof. Let $\alpha \in E$ be given. Then, there exist not all zero $a_0, \ldots, a_n \in K$ such that

$$a_n\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$$

Since each $a_i \in K$ is algebraic over F, thus $F(\alpha_0, \ldots, \alpha_n)$ is finite over F, being above theorem.
Further,

$$F(\alpha, a_0, \ldots, a_n) = F(a_0, \ldots, a_n)(\alpha)$$

is finite over $F(a_0, \ldots, a_n)$, because above equation gives that α is algebraic over $F(a_0, \ldots, a_n)$. Now,

$$[F(\alpha, a_0, \ldots, a_n) : F] = [F(\alpha, a_0, \ldots, a_n) : F(a_0, \ldots, a_n)][F(a_0, \ldots, a_n) : F]$$

is finite, thus algebraic over F. This means $\alpha \in E$ is algebraic over F. □

Theorem 0.0.17. Let E be an Extension field of F, and $\alpha \in E$.

$$F(\alpha) = \left\{ \frac{p(\alpha)}{q(\alpha)} \;\middle|\; p(x), q(x) \in F[x], q(\alpha) \neq 0 \right\}$$

0.0.3 Algebraic Closure

Lemma 0.0.18. Suppose that E is an Extension field of F. Then,

$$\overline{F}_E \stackrel{\text{def}}{=} \{\alpha \in E \mid \alpha \text{ is algebraic over } F\}$$

is a subfield of E. This $\overline{F}_E$ is called *Algebraic Closure* of F in E.

Proof. Let $\alpha, \beta \in \overline{F}_E$ be given. Since α is algebraic over F, thus $F(\alpha)$ is finite extension of F.
Then, $F(\alpha, \beta) = F(\alpha)(\beta)$ is finite over $F(\alpha)$, thus $F(\alpha, \beta)$ is finite extension over F, thus algebraic over F.
This implies $F(\alpha, \beta) \subseteq \overline{F}_E$. Now, with assumption $\beta \neq 0$,

$$\alpha + \beta, \alpha\beta, \frac{\alpha}{\beta} \in F(\alpha, \beta) \subseteq \overline{F}_E$$

□

Definition 0.0.19. A field F is said to be *Algebraically Closed* if:

Every non-constant polynomial in $F[x]$ has a zero in F

Theorem 0.0.20. A field F is Algebraically Closed *if and only if*

every non-constant polynomial in $F[x]$ factors into linear factors within $F[x]$.

Proof. Suppose that F is Algebraically Closed.
Let $f(x) \in F[x]$ be given. Since $f(x)$ has a root in F, put $\alpha_0 \in F$ such that $f(\alpha_0) = 0$. Then,

$$f(x) = (x - \alpha_0) f_1(x)$$

And, $f_1(x) \in F[x]$, it has a root in F, put $\alpha_1 \in F$ such that $f_1(\alpha_1) = 0$. Now,

$$f(x) = (x - \alpha_0)(x - \alpha_1) f_2(x)$$

In finite process, we get:

$$f(x) = c(x - \alpha_0)(x - \alpha_1) \cdots (x - \alpha_{\deg f(x)})$$

Conversely argument is clear. □

Lemma 0.0.21. Suppose that F is a field, and $f(x) \in F[x]$ is non-constant polynomial.
Then, there exists an Extension E such that $f(x)$ has a root in E.

Proof. Let $f(x) \in F[x]$. Since $F[x]$ is U.F.D., factors $f(x)$ into irreducibles in $F[x]$,

$$f(x) = f_1(x) \cdots f_n(x)$$

Now, for any $1 \leq j \leq n$, $F[x]/(f_j(x))$ is an Extension of F containing a root of $f(x)$. □

0.0.4 † Existence of Algebraic Closure

Definition 0.0.22. Suppose that F is a field. An Extension $\overline{F}$ is called *Algebraic Closure* of F if:

1. $\overline{F}$ is Algebraic over F.
2. Every nonconstant polynomial $f(x) \in F[x]$ factors completely into linear factors over $\overline{F}$.

Theorem 0.0.23. An Algebraic Clousre $\overline{F}$ is Algebraically Closed.

Proof. Lef $f(x) \in \overline{F}[x]$ be given. Then, there exists an Extension E of $\overline{F}$ containing a root $\alpha \in E$ of $f(x)$. Now, $\overline{F}(\alpha) \subset E$ is algebraic over $\overline{F}$, and by definition, $\overline{F}$ is algebraic over F. Thus $\overline{F}(\alpha)$ is algebraic over F. Particulary, α is algebraic over F, thus $\alpha \in \overline{F}$. □

Theorem 0.0.24. For any field F, there exists an Algebraically closed field K containing F.

Proof. Denote the set $M \overset{\text{def}}{=} \{f \in F[x] \mid \deg f \geq 1\}$. Set $S \overset{\text{def}}{=} \{x_f \mid f \in M\}$. Define the Polynomial Ring

$$F[S] \overset{\text{def}}{=} \bigcup_{\substack{\mathcal{F} \subset S \\ \mathcal{F} \text{ finite}}} F[\mathcal{F}]$$

Consider the ideal

$$I \overset{\text{def}}{=} (\{f(x_f) \mid f \in M\}) \subseteq F[S]$$

If $I = F[S]$, that is, I is Entire, then $1 \in I$. Thus, for some $g_1, \dots, g_n \in F[S]$ such that

$$g_1 f_1(x_{f_1}) + g_2 f_1(x_{f_2}) + \cdots + g_n f_n(x_{f_n}) = 1$$

For simplicity, write $x_i = x_{f_i}$. Since each g_i has only a finite number of variables, for $m \geq n$, write

$$\sum_{i=1}^{n} g_i(x_1, \dots, x_n, \dots, x_m) f_i(x_i) = 1$$

For each $1 \leq i \leq n$, let F_i be an Extension of F such that f_i has a root α_i.
Then, in the Compostion Field $F_1 \cdots F_n$,

$$1 = \sum_{i=1}^{n} g_i(x_1, \dots, x_n, \dots, x_m) f_i(\alpha_i) = 0$$

But this is Contradiction, thus I is Proper Ideal. The Zorn's Lemma gives the Existence of a Maximal Ideal

$$I \subsetneq \mathcal{M} \subsetneq F[S]$$

Thus, the Quotient ring

$$K_1 \overset{\text{def}}{=} F[S]/\mathcal{M}$$

is a Field containing F, and each polynomial f has a root in K_1.
Inductively, construct K_2 as an Extension of K_1 containing a root of all polynomial in K_1. We obtain

$$F = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_j \subseteq \cdots$$

Finally, define

$$K = \bigcup_{j=0}^{\infty} K_j$$

Then, K contains F, and any polynomial $k(x) \in K[x]$, for some $j \geq 0$, $k(x) \in K_j[x]$. And, $k(x)$ has a root in $K_{j+1}[x]$ by construction, thus K is Algebraically Closed. □

Theorem 0.0.25. Suppose that E is an Extension field of F, and Q is a Field of Fractions of $F[x]$. Then,

E is Not Algebraic over F *if and only if* There exists an Embedding $\varphi : Q \to E$ such that $\forall a \in F,\ \varphi(a) = a$.